Curriculum Vitae – Timothy Hnat, Ph.D.

Personal Information

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About Me

I have 18 years of experience developing cutting-edge AI technology for wearables and smart devices. Prior to becoming full-time CTO for CuesHub, I was the Chief Software Architect for two NIH-funded national research centers in AI for wearables where I led the development of research-grade software for wearables, smartphones, and cloud platforms that enabled cutting-edge scientific studies on sensor-triggered mobile interventions. Research groups in medicine, psychology, engineering, and computing across 11 states in the U.S. have collected hundreds of terabytes of wearable sensor data using this software to push the frontiers of research in a wide variety of health, wellness, and work performance domains. Previously, as a tenure-track Assistant Professor of Computer Science, my research resulted in new software platforms for smart homes and AI for wearables.

Key Skills and Technologies

Data Science & AI/ML: Machine Learning, Predictive modeling, LLMs, Statistical Analysis

Data Engineering: Signal Filtering, Python, PostgreSQL/MySQL, Apache Spark

Infrastructure: Docker, AWS, Bare Metal

Visualization Tools: Grafana, Kibana, Custom Portals, In-app (iOS and Android) data UIs

Education

University of Virginia Ph.D., Computer Science *2012*

University of Louisville M.Eng., Computer Engineering and Computer Science *2006*

University of Louisville B.S., Computer Engineering and Computer Science *2005*

Employment

Postdoctoral RD Entrepreneurship Fellow UofM and Epicenter *2025–present*

Patents2Product Post-Doc fellow supported by the University of Memphis and Epicenter to start a for-profit business utilizing patented intellectual property developed in Memphis.

Chief Technology Officer and Co-founder CuesHub, PBC *2022–present*

CuesHub is a deep-tech startup commercializing biometric AI to support mental health and stress resilience.

1. Implement an end-to-end data strategy spanning sensor integration, signal acquisition, model development, and deployment on wearable and mobile platforms.
2. Designed and built real-time machine learning pipelines for stress detection and adaptive intervention using multimodal data from smartwatches and smartphones.
3. Designed and deployed a privacy-preserving infrastructure to enable continuous, real-world data collection and personalized biometric analytics.

Chief Software Architect and Local Deputy Director University of Memphis *2020–2023*

mHealth Center for Discovery, Optimization & Translation
of Temporally-Precise Interventions (mDOT) **Total: $5.9 million**

1. Served as primary point of contact for interdisciplinary collaborators-including behavioral scientists, mathematicians, and clinical researchers-to scope, design, and deliver robust digital health solutions.
2. Led a cross-functional team of engineers and researchers in translating cutting-edge biomedical research into scalable, real-world mobile health studies.
3. Co-designed and supported the Mobile Open Observation of Daily Stressors (MOODS) project, enabling passive sensing and active ecological assessments in the field.
4. Designed, developed, and deployed the first real-time stress assessment app for Fossil smartwatches, integrating biometric signals into mobile interventions.
5. Led design and development of a personalized Just-in-Time Adaptive Intervention (pJITAI) engine to support precision behavioral science research.

6. Expanded the data processing framework to enhance analytic capabilities and improve integration with Jupyter Notebooks for interactive data exploration and visualization.
7. Contributed to the design of a digital toolbox for organizing and disseminating mHealth technologies, including datasets, frameworks, models, toolkits, and platforms.

Chief Software Architect University of Memphis *2014–2023*
NIH Center of Excellence for Mobile Sensor Data-to-Knowledge (MD2K) **Grants: $39.8 million**

1. Served as primary point of contact for interdisciplinary collaborators-including behavioral scientists, mathematicians, and clinical researchers-to scope, design, and deliver robust digital health solutions.
2. Led a cross-functional team of five engineers and researchers in operationalizing cutting-edge biomedical research into scalable, real-world studies (e.g., mProv, SES, AA, ROBAS, mPerf).
3. Designed and deployed a high-throughput data analytics infrastructure using Apache Spark and Python to support ingestion and processing of 300+ TB of multi-modal sensor and signal data from nationwide research participants.
4. Designed and implemented a custom data processing framework that remains integral to the Center's ongoing research workflows.

Assistant Professor University of Memphis *2011–2014*
Led data-driven research on indoor tracking and navigation systems, mobile health interventions, and body sensor networks – leveraging machine learning and real-time analytics to generate context-aware behavioral and clinical insights.

Graduate Student Researcher University of Virginia *2006–2012*
Conducted research on programming systems, domain-specific languages, and scalable data science and engineering techniques for large-scale wireless embedded sensor networks.

Teaching Assistant University of Virginia *2006–2008*

Dissertation

A System for Tracking People in Homes for Smart Home Applications
People spend approximately 62% of their time indoors, yet conventional localization technologies like GPS are unreliable in indoor environments. Accurate identification of both individuals and their room-level location is critical for enabling intelligent, context-aware smart home systems. This work presents a novel hardware-software solution for indoor tracking, combining sensor fusion, identity resolution, and spatial inference to enable precise, real-time localization within residential spaces.

Selected Projects

MOODS (University of Memphis) *2020–2023*

- The Mobile Open Observation of Daily Stressors (MOODS) study aims to identify key stress events in everyday life using multimodal sensor data. By analyzing physiological signals, behavioral patterns, and environmental context, the study seeks to uncover the complex interactions that contribute to stress. Findings from MOODS may advance our understanding of real-world stressor detection and inform the development of personalized, data-driven interventions for mental health and well-being.
- https://moods.md2k.org

mContain (University of Memphis) *2020*

- Researchers at the MD2K Center of Excellence, based at the University of Memphis, developed and launched mContain, a free mobile app designed to promote social distancing and reduce COVID-19 community transmission. mContain leverages smartphone GPS and Bluetooth technologies to detect proximity encounters-defined as close contact within six feet for several minutes-with other app users. The app provides users with a daily count of these encounters, similar to a step counter, to encourage awareness of social interactions. Additionally, mContain displays real-time crowding levels on a map, helping users avoid densely populated areas and make safer movement decisions.
- https://mcontain.md2k.org

mCerebrum (University of Memphis) *2014–2021*

- mCerebrum is a configurable software platform for mobile and wearable sensing that supports high-throughput, reliable data collection and real-time processing. It enables sensor-driven, just-in-time adaptive interventions (JITAIs) by allowing dynamic integration of diverse data streams and responsive in-the-moment feedback mechanisms.
- http://github.com/MD2Korg/

Cerebral Cortex (University of Memphis) *2014–Present*

- Cerebral Cortex is a modular big-data architecture (Apache Spark and Python) built on a flexible, layered design, enabling independent development and extensibility across functional components. A central Kernel orchestrates communication between layers, providing unified data access, enforcing security controls, and abstracting implementation details to ensure system scalability and maintainability.
- http://github.com/MD2Korg/

Smart home tracking (University of Virginia) *2009–2012*

- I developed a hardware and software system mounted above doorways to track individuals within their homes without requiring cameras or wearable devices. The system achieved over 90% tracking accuracy and generated more than 2TB of reliably logged data during deployment.
- Demo: https://www.youtube.com/watch?v=wAluI_uniK8

MacroLab (University of Virginia) *2007–2009*

- I developed a complete toolchain for running Matlab-like code on a distributed wireless sensor network, deploying and testing it on a 50-node testbed. This led to a 100x reduction in code size and the creation of a debugging environment to support the new programming abstraction.

Advisory Board

University of Memphis, Memphis, Tennessee

- Center for Information Assurance *2012–2015*

Awards and Honors

University of Memphis, Memphis, Tennessee

- NCURS demonstration presenter to 3,500 undergraduates with Santosh Kumar's keynote *2017*
- mHealth Scholar *2013*
- Ralph E. Powe Junior Faculty Enhancement Nomination *2013*

University of Virginia, Charlottesville, Virginia

- Frank Anger Memorial ACM SIGBED/SIGSOFT Student Award, *2009*
- SenSys Student Travel and Conference Funding Award, *2009*
- SenSys Student Conference Funding Award, *2008*
- IPSN Student Travel and Conference Funding Award, *2008*
- UVA Fellowship, *2006–2011*

University of Louisville, Louisville, Kentucky

- Fischer Family Scholarship, University of Louisville, *2001–2006*
- ACM Distinguished Student Award, *2005*
- Speed School Alumni (Scholarship), *2001–2002*

Publications

1. Neupane S., Saha M., ALi N., **Hnat, T.**, Samiei S.A., Nandugudi A., Almeida D.M., Kumar S. *Momentary Stressor Logging and Reflective Visualizations: Implications for Stress Management with Wearables.* arXiv preprint arXiv:2401.16307, 2024
2. Hojjatinia S., Daly E.R., **Hnat, T.**, Hossain S.M., Kumar S., Lagoa C.M., Nahum-Shani I., Samiei S.A., Spring B., Conroy D.E. *Dynamic System Modeling to Identify Moments of Vulnerability Based on Stress-Smoking Responses in Daily Smokers.* Annals of Behavioral Medicine. 56, 2022
3. Hojjatinia S., Daly E.R., **Hnat, T.**, Hossain S.M., Kumar S., Lagoa C.M., Nahum-Shani I., Samiei S.A., Spring B., Conroy D.E. *Dynamic models of stress-smoking responses based on high-frequency sensor data.* npj Digital Medicine. 4, 162 2021
4. Shetty V, Morrison D, Delin T, **Hnat T**, Kumar S, *A Scalable System for Passively Monitoring Oral Health Behaviors Using Electronic Toothbrushes in the Home Setting: Development and Feasibility Study*, JMIR Mhealth Uhealth, 2020

5. Yli-Piipari S, Zaman K, Fish J, **Hnat T**, *Validation of K-Sense against Indirect Calorimetry and Accelerometry*, Human Movement, pp. –, 2018
6. Hossain SM, **Hnat T**, Saleheen N, Nasrin NJ, Noor J, Ho B-J, Condie T, Srivastava M and Kumar S, *mCerebrum: An mHealth Software Platform for Development and Validation of Digital Biomarkers and Interventions*, Proceedings of the 15th ACM Conference on Embedded Network Sensor Systems (SenSys), Delft, Netherlands, pp. –, November 2017
7. **Hnat T**, Hossain S, Ali N, Carini S, Condie T, Sim I, Srivastava M and Kumar S, *mCerebrum and Cerebral Cortex: A Real-time Collection, Analytic, and Intervention Platform for High-frequency Mobile Sensor Data*, In AMIA (American Medical Informatics Association) 2017 Annual Symposium., November 2017.
8. Kumar S, Abowd G, Abraham WT, al'Absi M, Chau DH, Ertin E, Estrin D, Ganesan D, **Hnat T**, Hossain SM, Ives Z, Kerr J, Marlin BM, Murphy S, Rehg JM, Nahum-Shani I, Shetty V, Sim I, Spring B, Srivastava M and Wetter D, *Center of Excellence for Mobile Sensor Data-to-Knowledge (MD2K)*, IEEE Pervasive Computing., Vol. 16(2), pp. 18-22., April 2017
9. Kazi I. Zaman, Anthony White, Sami Yli-Piipari, **Timothy W. Hnat** *K-Sense: A Kinematic Approach to Measuring Human Energy Expenditure for Daily Living Activities.* Proceedings of The 11th European Conference on Wireless Sensor Networks (EWSN), Oxford, United Kingdom, pp. –, February 2014
10. **Timothy W. Hnat**, Erin Griffiths, Raymond Dawson, Kamin Whitehouse. *Doorjamb: Unobtrusive Room-level Tracking of People in Homes using Doorway Sensors.* Proceedings of the 10th ACM Conference on Embedded Network Sensor Systems (SenSys), Toronto, Canada, pp. –, November 2012
11. **Timothy W. Hnat**, Vijay Srinivasan, Jiakang Lu, Tamim Sookoor, Raymond Dawson, John Stankovic, Kamin Whitehouse. *The Hitchhiker's Guide to Successful Residential Sensing Deployments.* Proceedings of the 9th ACM Conference on Embedded Network Sensor Systems (SenSys), Seattle, WA, pp. –, November 2011
12. **Timothy W. Hnat**, Kamin Whitehouse. *A Relaxed Synchronization Primitive for Macroprogramming Systems.* Proceedings of the 7th International IEEE Conference on Networked Sensing Systems (INSS), Kassel, Germany, pp. 219–226, June 2010
13. **Timothy W. Hnat**, Tamim I. Sookoor, Pieter Hoomimeijer, Westley Weimer, Kamin Whitehouse. *A Modular and Extensible Macroprogramming Compiler.* Proceedings of the 1th Workshop on Software Engineering for Sensor Network Applications (SESENA) in Conjunction With ACM/IEEE International Conference on Software Engineering (ICSE), Cape Town, South Africa, pp. 49–54, May 2010
14. Tamim I. Sookoor, **Timothy W. Hnat**, Pieter Hoomimeijer, Westley Weimer, Kamin Whitehouse. *Macrodebugging: Providing Abstract Views of System State.* Proceedings of the 7th ACM Conference on Embedded Network Sensor Systems (SenSys), Berkeley, CA, pp. 141–154, November 2009
15. **Timothy W. Hnat**, Tamim I. Sookoor, Pieter Hoomimeijer, Westley Weimer, Kamin Whitehouse. *MacroLab: A Vector-based Macroprogramming Framework for Cyber-Physical Systems.* Proceedings of the 6th ACM Conference on Embedded Network Sensor Systems (SenSys), Raleigh, NC, pp. 225–238, November 2008
16. S. Braun, W. P. Hnat, **T. W. Hnat**, H. L. Legan, *Taking the guesswork out of mandibular symphyseal distraction osteogenesis.* American Journal of Orthodontics and Dentofacial Orthopedics, Volume 119, Number 2, pp. 121–126, February 2001
17. S. Braun, W. P. Hnat, B. Kusnoto, **T. W. Hnat**, *A new accurate approach to the anterior ratio with clinical applications. Part 1: A computer program.* American Journal of Orthodontics and Dentofacial Orthopedics, Volume 115, Number 4, pp. 368–372, April 1999

Conference Demos

1. R. Dickerson, **T. Hnat**, E. Hoque, J. Stankovic. *Demonstration of Sleep Monitoring and Caregiver Displays for Depression Monitoring* . Wireless Health, San Diego, CA, October 2011
2. **Timothy W. Hnat**, Tamim I. Sookoor, Kamin Whitehouse. *Macrodebugging with MDB Framework for Cyber-Physical Systems.* The 7th ACM Conference on Embedded Network Sensor Systems (SenSys), Berkeley, CA, November 2009

3. Tamim I. Sookoor, **Timothy W. Hnat**, Kamin Whitehouse. *Demo Abstract: Programming Cyber-Physical Systems with MacroLab.* The 6th ACM Conference on Embedded Network Sensor Systems (SenSys), Raleigh, NC, November 2008
4. R. Dickerson, J. Lu, B. Chantree, **Timothy W. Hnat**, J. Lu, J. Stankovic, K. Whitehouse, *MetroNet: Case Study for Collaborative Data Sharing on the World Wide Web.* Information Processing and Sensor Networks, April 2008